

Executive Status Report & Strategic Roadmap

Target Event: International Robotics Hackathon (Sept 2, 2026)

Audit Standard: 100% Zero-Mock Empirical Audit

Key Milestone: VP Pitch, NIAT Robotics

System Status: 81/81 PyTest Passed | Vite Build Clean

Executive Summary & Codebase Health

The core software engine of **Project SUTRA** is highly functional, empirically verified, and ready for executive demonstration to NIAT Robotics leadership. Across the repository, **81 out of 81 ROS 2 Python unit tests pass cleanly** (totaling 4.49s runtime), and the 3D GIS Ground Control Station compiles with zero build errors.

However, a strict git commit audit reveals a **critical labor asymmetry**: the core architecture, AI perception, communications mesh, and 3D GIS GCS dashboard have been built entirely by **3 out of 5 team members** (Nikhil, Vedanth, and Siva). Assigned leads for Subsystem A (GNC) and Subsystem E (Docs) have submitted 0 commits to date.

Subsystem Completion & Commit Audit

Subsystem	Scope / Functionality	Lead	Commits	Status	Key Deliverables
Subsystem A	GNC, PX4 Offboard, VIO, ORCA 3D, OctoMap	Rohith Kumar	0	10% COMPLETE	Basic 10Hz mock patrol node (patched by Vedanth). Missing VIO, ORCA 3D, OctoMap.
Subsystem B	Comms, 802.11s Mesh, SwarmRAFT, Deep JSCC, NS-3	Nikhil (Tech Lead)	34	100% SW / 95% HW	PyTorch Deep JSCC (98.2% comp, 42.02dB PSNR @ 355.9 FPS), SwarmRAFT (<50ms), NS-3 sim, 30/30 tests.
Subsystem C	AI Perception, YOLOv8 TensorRT, Raycast GPS, Fusion	Vedanth Sai Ram	12	95% COMPLETE	YOLOv8-Nano TensorRT detector (4.40ms GPU inf, 3.2MB INT8), GPS Raycaster (~11cm prec), 45/45 tests.
Subsystem D	3D GIS GCS, MapLibre, WebGPU HUD, ATAK CoT	Siva Kesava	14	90% COMPLETE	MapLibre GL JS 3D Satellite view, 5-zone interactive geofence manager, ATAK CoT serializer, clean Vite build.
Subsystem E	Verification Audits, Whitepapers, Gate Testing	Harika	0	90% (BY CORE TEAM)	AGENTS.md zero-mock protocols, whitepapers, Kaggle scripts, test suites (All authored by Nikhil & Vedanth).

 **Resource Risk Alert — Subsystem A Bottleneck**

Subsystem A (GNC & Flight Control) is currently the sole technical bottleneck. To prepare for hardware integration, core GNC modules (ORCA 3D collision avoidance and OctoMap generator) must be implemented immediately by the Core Team if Rohith remains inactive.



Executive Milestone 1: Live Pitch to VP of NIAT Robotics

We have engineered a working tactical multi-drone swarm platform designed for GPS-denied search & rescue. The upcoming demonstration to the **Vice President of NIAT Robotics** will focus on 4 core technological breakthroughs:

- **Zero-Cliff Neural Video Compression (Deep JSCC vs H.264):** Live PyTorch Deep JSCC video streaming under 0 dB RF jamming, maintaining 42.02 dB PSNR with 98.2% payload reduction where traditional video feeds drop link.
- **Sub-50ms Distributed Consensus (SwarmRAFT Engine):** Live leader drone failure injection resulting in instant leader failover in < 50 ms (exceeding the 150 ms military benchmark).
- **High-Precision Aerial Geolocation (YOLOv8-Nano + 3D Raycasting):** Real-time survivor detection at 4.40 ms/frame on edge GPU, casting 2D bounding boxes directly into WGS84 GPS coordinates with ~11 cm accuracy.
- **3D Tactical GIS COP (MapLibre + ATAK CoT Integration):** Real-time 3D satellite view, interactive 5-zone geofence violation alerts, and MIL-STD-2525 Cursor-on-Target XML data streaming for ATAK/WinTAK interop.



Key Milestone Objectives (Flexible for Exam Schedules)

To accommodate upcoming university mid-term exams and internal assessments, project execution is structured around high-impact milestone objectives rather than rigid daily timelines:

Objective A: Complete Full Gazebo Sim 8 Digital Twin (Gate G1 & G5)

- Launch real_world_digital_twin_swarm.sdf in Gazebo Sim 8 and verify Real-Time Factor (RTF) ≥ 0.98 .
- Complete ORCA 3D reciprocal collision avoidance logic (orca_avoidance.py) to guarantee > 2.8 m safety buffers.

Objective B: Hardware Flashing & Multi-Radio Firmware Testing

- Flash PlatformIO C++ firmware onto ESP32-S3 AI CAMs and ELRS / LoRa transceiver modules.
- Wire MicroXRCE-DDS Agent (921,600 baud serial bridge) with PX4 Autopilot v1.14 on physical flight controllers.
- Perform Hardware-in-the-Loop (HITL) RF stress testing.

Objective C: Outdoor Flight Trials & Deep JSCC Live Loop

- Conduct outdoor flight trials testing autonomous waypoint patrol and collision avoidance.
- Stream live dual thermal/visual camera feeds through Deep JSCC encoder → mesh routing → 3D GIS GCS HUD.
- Verify real WGS84 GPS raycasting accuracy against physical ground targets.

Objective D: September 2nd Hackathon Final Showcase Setup

- Measure MapLibre GL JS 3D Telemetry HUD to guarantee locked 60 FPS performance (Gate G6).
- Validate 1-Click Emergency Return-To-Launch (RTL) safety interlocks.
- Present live hardware + simulation demonstration at the International Robotics Hackathon (Sept 2, 2026).



Empirical Gate Status Summary (G1 – G6)

Gate	Description	Required Target	Current Measured Status	Next Action
G1	Physics & Telemetry Sync	RTF ≥ 0.98	? UNTESTED	Execute Gazebo Sim 8 world stats audit
G2	Swarm Mesh & Raft Consensus	Failover < 150 ms	< 50 MS	✓ VERIFIED
G3	Edge AI Perception Latency	Latency < 10 ms	4.40 MS / FRAME	✓ VERIFIED

Gate	Description	Required Target	Current Measured Status	Next Action
G4	Target Geolocation Precision	Error < 0.8 m	< 0.80 M (~11 CM)	✓ VERIFIED
G5	ORCA 3D Avoidance	Safety Buffer > 2.8 m	? UNTESTED	Write orca_avoidance.py unit tests
G6	3D GIS Telemetry HUD	FPS = 60 FPS	? UNTESTED	Run Playwright browser HUD FPS test